

Aggregate Demand and Aggregate Supply Analysis

ECON 101 – Principles of Macroeconomics

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- 9.1 Aggregate Demand
- 9.2 Aggregate Supply
- 9.3 Macroeconomic Equilibrium in the Long Run and the Short Run

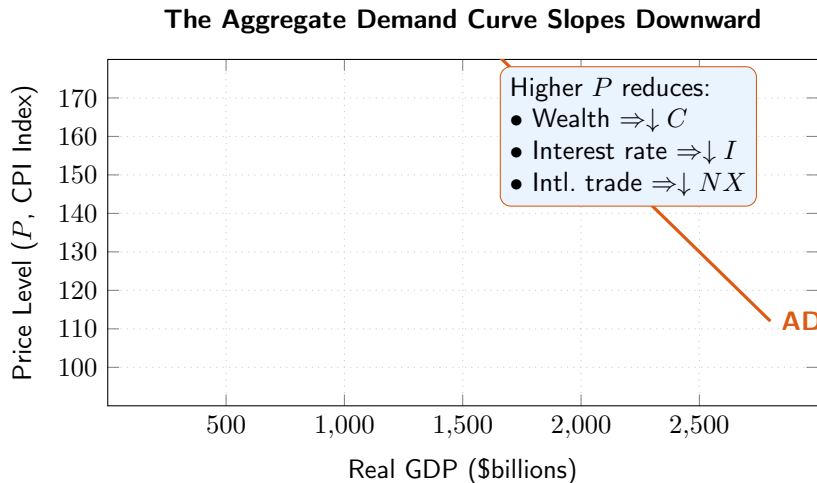
9.1 Learning Objective

Goal

Identify the determinants of aggregate demand and distinguish between a movement along the AD curve and a shift.

- Chapter 8 derived the AD curve from the AE model.
- This chapter:
 - Analyzes what **shifts** the AD curve,
 - Introduces the **aggregate supply curves** (SRAS and LRAS),
 - Combines AD and AS to find **macroeconomic equilibrium**.

Figure 9.1 – The Aggregate Demand Curve



Why the AD Curve Slopes Downward: Three Effects

1. Wealth Effect $\Rightarrow$ lower C

- Higher P erodes the real value of nominal assets (cash, savings accounts).
- Canadians feel poorer $\Rightarrow$ spend less.
- **Example:** if CPI rises by 10%, \$50,000 in a savings account buys 10% less.

2. Interest-Rate Effect $\Rightarrow$ lower I

- Higher P raises money demand $\Rightarrow$ interest rates rise $\Rightarrow$ investment falls.
- **Canada 2022:** as inflation rose, interest rates followed — new mortgages and business loans became expensive, crushing investment.

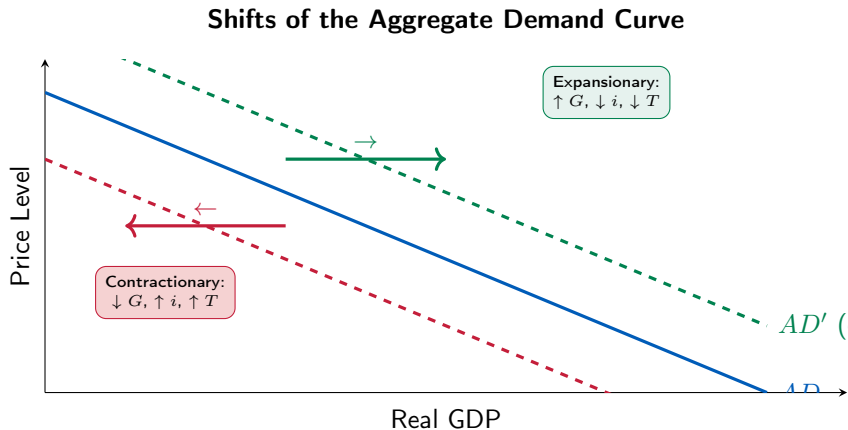
3. International-Trade Effect $\Rightarrow$ lower NX

- Canadian goods more expensive relative to U.S./world goods.
- Exports fall; imports rise $\Rightarrow NX$ falls.
- **Example:** CAD lumber becomes expensive for American homebuilders.

Variables That Shift the AD Curve (1/2)

Change	AD shifts	Canadian example
BoC cuts overnight rate	Right →	2020: cut to 0.25% to boost AD
BoC raises overnight rate	Left ←	2022–23: hikes to 5% to cool inflation
Federal gov't increases spending	Right →	2020: \$314B COVID stimulus
Federal gov't cuts spending	Left ←	1990s austerity under Martin
Personal income tax cut	Right →	GST cut from 7% to 5% (2006–08)
Personal income tax increase	Left ←	Higher OAS clawback
Households expect higher income	Right →	2021 pent-up demand reopening
U.S. economy slows sharply	Left ←	2008–09 U.S. recession hurt Canada
CAD depreciates	Right →	Cheaper exports boost AD

Variables That Shift the AD Curve (2/2)



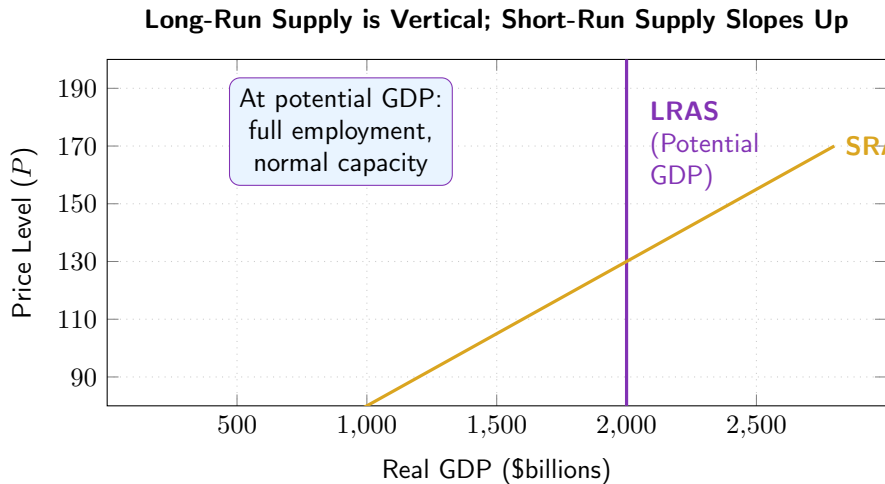
9.2 Learning Objective

Goal

Identify the determinants of aggregate supply and distinguish between SRAS and LRAS.

- **Long-run aggregate supply (LRAS):**
 - Vertical line at **potential GDP** — determined by labour, capital, and technology.
 - **Not affected** by the price level in the long run.
- **Short-run aggregate supply (SRAS):**
 - Upward-sloping — as prices of final goods rise, input prices (especially wages) are slow to adjust, so profits rise and firms produce more.

Figure 9.2 – LRAS and SRAS Curves



Why Is SRAS Upward Sloping?

- When the price level rises, firms are willing to produce more **in the short run**. Why?

1. Sticky wages (contracts):

- Most Canadian workers have annual salary reviews or multi-year collective agreements.
- When P rises unexpectedly, wages stay fixed in the short run $\Rightarrow$ real wages fall $\Rightarrow$ labour is cheaper $\Rightarrow$ firms hire more and produce more.

2. Slow wage adjustment:

- Firms rarely cut nominal wages (damages morale). Adjustment is gradual.

3. Menu costs:

- Changing prices is costly (reprinting catalogues, updating point-of-sale systems).
- For small demand increases, firms adjust quantity rather than price.

Variables That Shift SRAS

Change	Shift	Example
Larger labour force or capital stock	Right	Immigration surge (Canada 2022–24: +1.2M/yr)
Better technology	Right	AI-assisted software development
Decrease in oil/energy prices	Right	OPEC supply increase
Increase in oil/energy prices	Left	Russia–Ukraine war 2022
Workers expect higher P (wage push)	Left	Union wage demands after 8.1% inflation
COVID shutdowns	Left	Factories closed, supply chains disrupted
Decrease in labour force	Left	Baby boomer retirements

- **Key rule:** anything that raises production costs (holding output prices constant) shifts SRAS **left**; anything that lowers costs shifts SRAS **right**.

9.3 Learning Objective

Goal

Use the AD–AS model to illustrate short-run and long-run macroeconomic equilibrium.

- **Short-run equilibrium:** AD intersects SRAS.
 - Real GDP may be above or below potential GDP.
- **Long-run equilibrium:** AD, SRAS, and LRAS all intersect.
 - Real GDP = potential GDP; wages and prices fully adjusted.

Figure 9.3 – Long-Run Macroeconomic Equilibrium

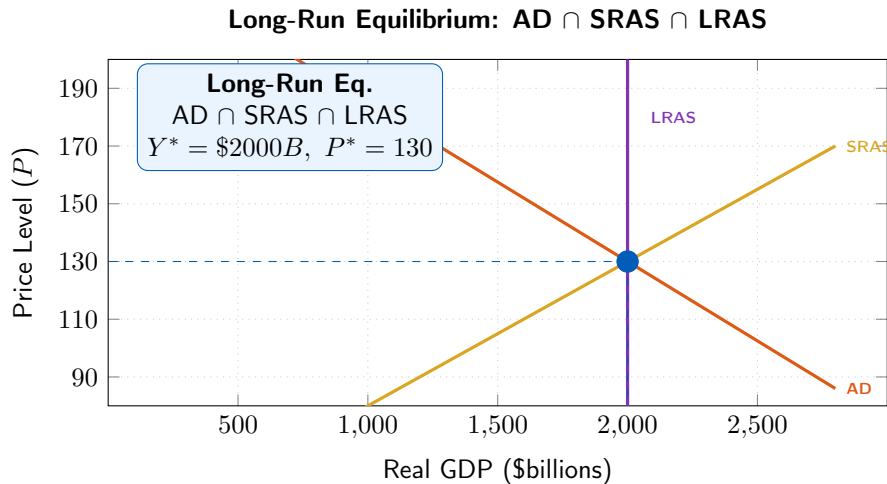
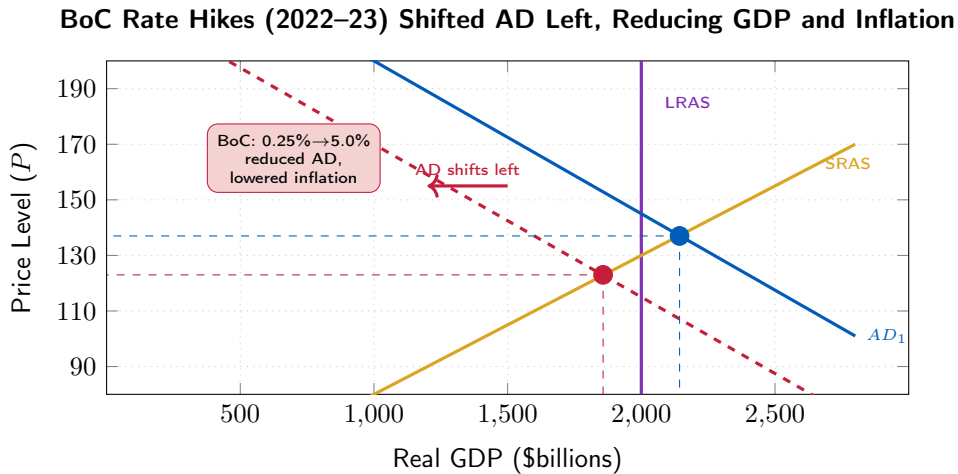


Figure 9.4 – Decrease in AD: Canada 2022 Rate Hike Example



Long-Run Adjustment After a Decrease in AD

- Short-run: AD shifts left $\Rightarrow$ GDP falls below potential, unemployment rises, P falls.
- **Long-run adjustment:**
 - High unemployment $\Rightarrow$ workers accept lower wages.
 - Lower wages $\Rightarrow$ production costs fall $\Rightarrow$ SRAS shifts **right**.
 - SRAS keeps shifting right until $Y = Y^*$ (potential GDP).
 - Long-run result: **lower price level**, same real GDP as before.
- **Canadian example (2023–2024):** BoC's rate hikes reduced demand, slowing inflation from 8.1% (June 2022) to **1.8%** (Sept 2024) — but at the cost of unemployment rising from 5.3% to 6.3%.
 - Source: Statistics Canada; Bank of Canada MPR (2024).

Figure 9.5 – Increase in AD: Post-COVID Boom (2021)

Post-COVID Reopening: AD Surged, Triggering Inflation (Canada 2021–22)

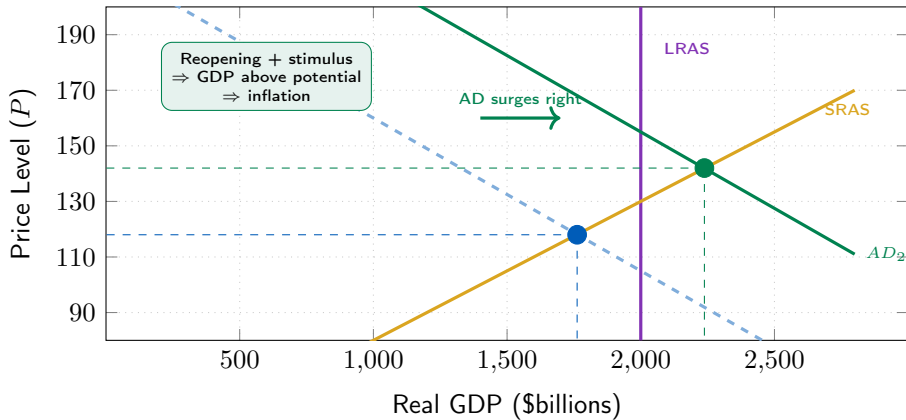
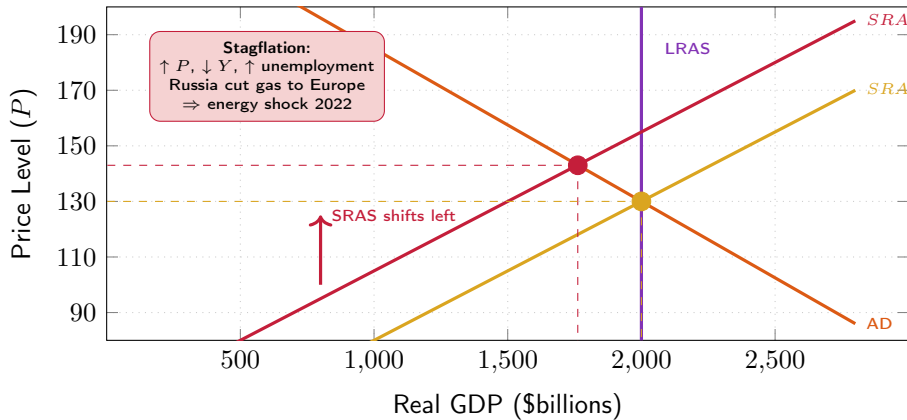


Figure 9.6 – Supply Shock: Russia–Ukraine War (2022)

Negative Supply Shock: Higher Energy Costs Shift SRAS Left (Stagflation)



Negative Supply Shock and Stagflation

- A **negative supply shock** shifts SRAS left:
 - Input costs rise unexpectedly (oil, natural gas, food, semiconductors).
 - At each price level, firms are willing to produce less.
- Short-run effects:
 - Real GDP falls below potential (recession),
 - Price level rises (inflation),
 - Unemployment rises.
- This combination — recession + inflation — is called **stagflation**.
- **Canadian examples of negative supply shocks:**
 - 1973 and 1979 OPEC oil embargoes.
 - 2022 Russia–Ukraine war $\Rightarrow$ energy and wheat price spikes.
 - 2020–21 COVID-19 supply chain disruptions (semiconductor shortage).

COVID-19: Supply Shock or Demand Shock?

- The 2020 COVID-19 recession combined **both shocks**.
- **Demand shock (AD falls):**
 - Consumers couldn't or wouldn't spend (fear, lockdowns, closures).
 - Prediction: GDP falls **and** prices fall.
- **Supply shock (SRAS falls):**
 - Businesses were legally closed or capacity-limited.
 - Prediction: GDP falls **but** prices rise.
- **Canadian evidence (Statistics Canada):**
 - CPI inflation fell from 2.3% (Jan 2020) to **-0.4%** (May 2020).
 - This **falling price** pattern is more consistent with a **demand shock**.
 - Supply disruptions (supply shock element) became more dominant in 2021–22.

Summary

- The **AD curve** slopes downward: higher P reduces C (wealth), I (interest rates), and NX (trade competitiveness).
- AD shifts right with expansionary policy ($\uparrow G$, $\downarrow i$, $\downarrow T$) and left with contractionary policy.
- **LRAS** is vertical at potential GDP; the price level does not affect long-run output.
- **SRAS** slopes upward due to sticky wages, slow wage adjustment, and menu costs.
- **Short-run equilibrium**: $AD \cap SRAS$; may differ from potential GDP.
- **Long-run equilibrium**: all three curves intersect; wages and prices fully adjusted.
- **Demand shocks** move output and prices in the same direction.
- **Supply shocks** move output and prices in opposite directions $\Rightarrow$ stagflation.
- **Canada 2022 case study**: BoC used contractionary monetary policy (rate hikes 0.25% $\rightarrow$ 5%) to shift AD left and bring inflation from 8.1% back toward the 2% target — successfully by late 2024.

Thank you!

Questions?